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REVIEW

Global Prevalence of Social Anxiety Disorder in Children, Adolescents and Youth: A Systematic Review and Meta-analysis

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Abstract
Social anxiety disorder is a prevalent mental health condition that significantly impairs social interactions, academic performance, and professional functioning in children, adolescents, and youth. This study aimed to investigate the global prevalence of social anxiety disorder across these developmental stages. Six electronic databases (PubMed, Scopus, Web of Science, Embase, ScienceDirect and Google Scholar) were systematically searched for studies related to the prevalence of social anxiety disorder in children, adolescents and youth. Random-effects models were employed for data analysis and Comprehensive Meta-Analysis Version 2.0. Heterogeneity was assessed using the I^2 index. A total of 38 studies were included in the final analysis. The global prevalence of social anxiety disorder was estimated to be 4.7% in children, 8.3% in adolescents, and 17% in youth. These findings suggest a progressive increase in the prevalence of SAD across these developmental stages. Considering the prevalence of social anxiety disorder in this study, policymakers can use the findings of this study to inform and develop effective prevention strategies for individuals and communities most susceptible to this disorder.

Keywords Social anxiety disorder · Children · Adolescent · Meta-analysis

Background

Anxiety disorders are a class of mental disorders affecting children, adolescents and adults. The Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5), identifies several types, including generalized anxiety disorder, separation anxiety, panic disorder, specific phobias, and social anxiety disorder (Krygsman & Vaillancourt, 2022; Mpairwe et al., 2021; Nakie et al., 2022). Among these, social anxiety disorder is recognized as the third most common mental health disorder

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globally and the most frequent anxiety disorder among teenagers (Krygsman & Vaillancourt, 2022; Mpairwe et al., 2021; Nakie et al., 2022).

Social anxiety disorder is a chronic mental illness characterized by excessive fear in social situations and preoccupation with negative evaluation by others (Mohammadi et al., 2020). Symptoms persist for at least 6 months, often with onset in childhood or adolescence and can continue into adulthood (Garcia & O'Neil, 2021; Torabi et al., 2023). Individuals with social anxiety disorder experience difficulty interacting with or speaking to others, fearing rejection. In children, this may manifest as crying or screaming (Gilboa-Schechtman et al., 2020; Jefferies & Ungar, 2020; Torabi et al., 2023).

Several factors contribute to the development of social anxiety disorder in children and adolescents. Genetics, childhood temperament, parental loss, parental mental health, and physical, sexual or emotional abuse all increase the risk for social anxiety disorder. Parental employment status, education level and living arrangements (for example, independent living) are also risk factors (Al-Yateem et al., 2020).

Reports suggest a potential rise in the prevalence of social anxiety disorder (Jefferies & Ungar, 2020). However, there are significant inconsistencies across studies. For instance, one study reported a global prevalence ranging from 22.9 to 57.6%, while others found a 12-month prevalence of 0.8% in Europe and 0.2% in China (Jefferies & Ungar, 2020; Nakie et al., 2022). Lifetime prevalence in the United States and Canada varies between 8.1 and 12.1% (Jefferies & Ungar, 2020; Krygsman & Vaillancourt, 2022). The prevalence of social anxiety disorder appears to be particularly high among children entering adolescence (Altwaijri et al., 2023; Ernst et al., 2022; Madasu et al., 2019a; Shafi et al., 2023).

Studies have shown that fear of social interactions in young people with social anxiety disorder can lead to impaired academic performance and increase the risk of suicide and other mental health problems (Ernst et al., 2022). Individuals with social anxiety disorder also experience difficulties establishing social connections and face disruptions in personal and professional functioning (Gilboa-Schechtman et al., 2020; Jefferies & Ungar, 2020).

Given the concerning prevalence of social anxiety disorder among children, teenagers, and young people, coupled with the severe consequences of this disorder, effective preventive, control, and diagnostic measures are crucial. Accurate age-specific symptoms, behavioural changes and reliable prevalence data are essential for intervention strategies and policymaking. This study aims to address the inconsistencies in prevalence data by conducting a comprehensive review of existing literature to compile a more accurate global estimate of social anxiety disorder prevalence among children, adolescents, and youth.

Methods

This meta-analysis and systematic review adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Abdoli et al., 2022). A systematic search strategy was conducted on August 18, 2023, using six electronic databases: PubMed, ScienceDirect, Web of Science, Scopus, Embase, and Google Scholar. No time limit was applied during the initial search. The search was updated again in late August 2023. The following keywords were employed in various combinations using Boolean operators (AND and OR): prevalence, outbreak, burden, social anxiety disorder (SAD), social phobia, child, children, adolescent, teen, teenager, youth, and youth.

The World Health Organization (WHO) defines youth as 15–24-year-olds, adolescents as 10–19-year-olds, and children below 10 years old (Altwaijri et al., 2023). However, studies may utilize different age ranges (Jefferies & Ungar, 2020; Nakie et al., 2022). We aimed to categorize studies based on the WHO definitions and age groups reported in the studies themselves.

PubMed search string: (((((((((((Prevalence[Title/Abstract]) OR (outbreak[Title/Abstract])) OR (Burden[Title/Abstract])) AND (social anxiety disorder[Title/Abstract])) OR (SAD[Title/Abstract])) OR (social phobia[Title/Abstract])) AND (child[Title/Abstract])) OR (children[Title/Abstract])) AND (adolescent[Title/Abstract])) OR (teen[Title/Abstract])) OR (teenager[Title/Abstract])) OR (youths[Title/Abstract])) OR (young adults[Title/Abstract])))))))))).

Inclusions and Exclusion Criteria

Inclusion criteria: (1) Studies reporting the prevalence of social anxiety disorder in children, adolescents, and youth, either overall or stratified by age group. (2) Studies providing sufficient data, including sample size and prevalence rates. (3) Articles published in English or with English abstracts. (4) Studies that are based on the World Health Organization (WHO) age definitions for children (under 10 years), adolescents (10–19 years) and youth (15–24 years) (Altwaijri et al., 2023).

Exclusion criteria: (1) animal studies. (2) intervention studies, case reports, case series, and review articles. (3) duplicate publications. (4) Studies with participant age groups falling outside the defined ranges for children, adolescents and youth.

Choice of Study

Following the search strategy, retrieved studies were imported into Endnote software. Duplicates were removed, and to minimize bias, two reviewers independently screened titles and abstracts of remaining articles with blinded authorship and journal affiliation. Studies not meeting the inclusion criteria (outlined above) were excluded at this stage. Both reviewers then independently assessed the full text of the remaining articles for eligibility based on the pre-defined criteria. In cases where the opinions of the two researchers in charge of the review process differed, a third reviewer was asked to resolve the disagreement.

Quality Evaluation

The methodological quality of studies was evaluated using the 6-item Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) checklist, a common tool in review articles (Salari et al., 2020). The checklist assesses 6 domains (title, abstract, introduction, methodology, results, and discussion) with 32 questions. Studies scoring 16 or above were considered high-quality and included, while those scoring below 16 were excluded.

Data Extraction

A pre-designed checklist was used to extract data from included studies. This checklist captured information such as author name, publication year, location, sample size, age group, social anxiety disorder measurement tool, and prevalence data (number and percentage of individuals with social anxiety disorder). Two researchers independently conducted the data extraction process.

Statistical Analysis

Data analysis was performed using Comprehensive Meta-Analysis Version 2.0 software. Publication bias was assessed using Egger's test at a significance level of 0.05 and visually evaluated with a funnel plot. Heterogeneity among studies was assessed using the I^2 statistic.

Results

This systematic review and meta-analysis investigated the global prevalence of social anxiety disorder in children, adolescents and youth. PRISMA guidelines were followed to identify relevant studies without time restrictions (Fig. 1). A comprehensive search across databases yielded 1496 articles. Eight additional studies were identified through manual searching. After removing 377 articles due to duplication, the remaining studies (n = 1127) underwent title and abstract screening. Based on the pre-defined inclusion and exclusion criteria, 1007 articles were excluded, leaving 120 for full-text review. Following full-text assessment, 70 additional studies were excluded. Quality evaluation using the STROBE checklist resulted in the exclusion of 12 further studies. A total of 38 articles were included in the final analysis (Fig. 1). Details of these studies are presented in Tables 1, 2 and 3.

Table 1 summarizes studies on social anxiety disorder in children. Hussong et al. (2022) reported the highest prevalence (15.1%) among children aged 4.4 to 7.8 years using the DISYPS-III tool, while Parvaresh et al. (2015) reported the lowest rate of prevalence (0%) using K-SADS-PL among children aged 5 to 15 years (Hussong et al., 2022; Parvaresh et al., 2015). The overall pooled prevalence of social anxiety disorder in children in this study was 4.7%.

Table 2 presents studies on social anxiety disorder prevalence in adolescents. Capriola et al. (2017) reported the highest prevalence (45%) using the BNFE instrument among adolescents with an average age of 14.62 years. Mohammadi et al. (2020) reported the lowest prevalence (2.1%) using K-SADS-PL among

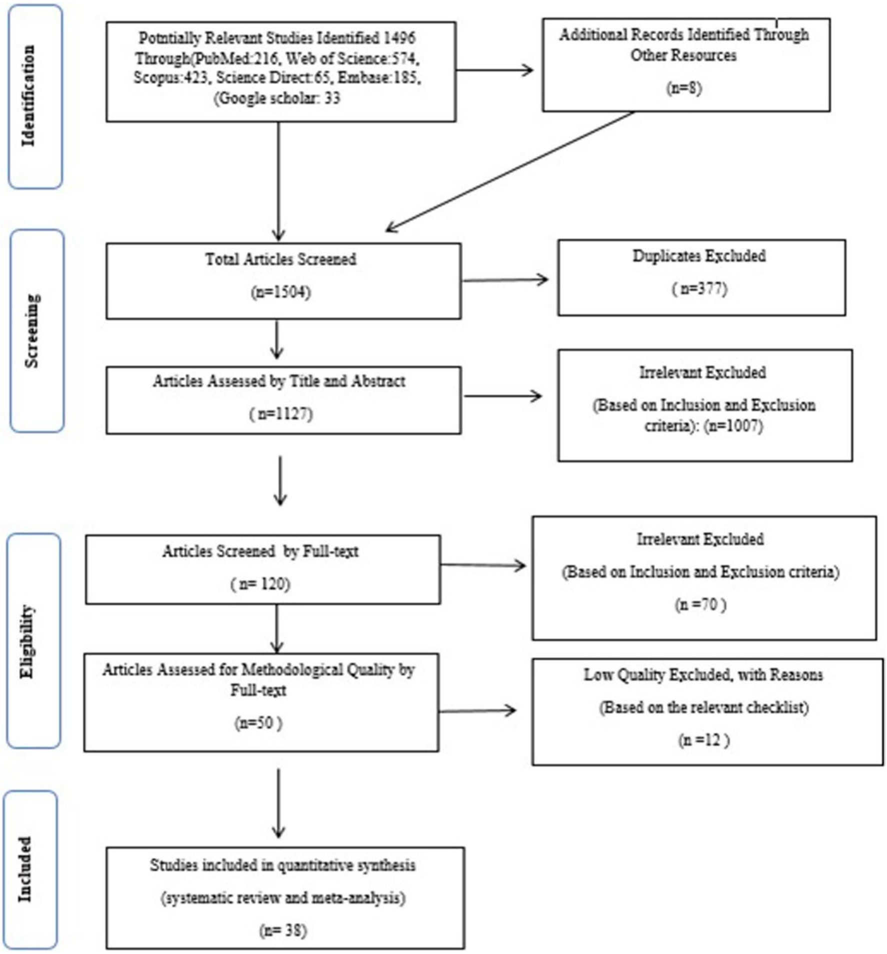


Fig. 1 The flowchart on the stages of including the studies in the systematic review and meta-analysis (PRISMA 2009)

adolescents aged 10–18 years. The overall pooled prevalence of SAD in adolescents in this study was 8.3%.

Table 3 presents studies on social anxiety disorder prevalence in youth. Uzonwanne (2014) reported the highest prevalence (95.6%) among young women and men aged 37.95 ± 12.19 and 37.35 ± 11.88, respectively. Mondin et al. (2013) reported the lowest prevalence (4%) using the MINI tool among young Brazilians aged 18 to 24 years. The overall prevalence of social anxiety disorder in youths in this study was 17%.

Table 1 Summary of characteristics of included studies of prevalence of social anxiety disorder in children

References	Location	Sample	SAD prevalence (%)	Age	Instrument
Abbo et al. (2013)	Uganda	702	5.30	3–9	M.I.N.I.-KID[a]
Parvaresh et al. (2015)	Iran	80	0	5–15	K-SADS[b]
Bener et al. (2011)	Qatar	610	10.20	6–11	Anonymous questionnaire and diagnoses with psychiatrists
Dodangi et al. (2014)	Iran	145	8.30	6–11	K-SADS-PL
Iverach et al. (2016)	Australia	150	9.34	7–12	YODA[c]
Ezpeleta et al. (2014)	Spain	622	1.90	3.8 ± 0.33	DICA-PPC[d]
Van Roy et al. (2009)	Norway	14,497	2.30	8–13	DAWBA[e]
Spence et al. (2018)	Australia	3334	1.60	4–11	DISC-IV[f]
Paulus et al. (2015)	Germany	1342	10.70	4–7	DISYPS-II[g]
Hussong et al. (2022)	Germany	1052	15.10	4.4–7.8	DISYPS-III
Mohammadi et al. (2020)	Iran	10,187	1.70	6–9	K-SADS-PL

[a]The MINI international neuropsychiatric interview for children and adolescents

[b]Kiddie-schedule for affective disorders and schizophrenia

[c]Youth Online diagnostic assessment

[d]Diagnostic interview of children and adolescents for parents of preschool children

[E]THE developmental and well being assessment

[F]THE Diagnostic interview schedule for children–version IV

[g]Diagnostik-system fur psychische storungen nach ICD-10 und DSM-IV fur kinder und jugendliche-II

Children

Eleven studies encompassing 32,721 participants were included in the analysis of social anxiety disorder prevalence among children. The I^2 heterogeneity test indicated high heterogeneity ($I^2 = 98.7$), prompting the use of the random-effects model for analysis. The pooled global prevalence of social anxiety disorders in children was estimated to be 4.7% (95% CI 2.5–8.4) (Fig. 2). The Egger test did not detect publication bias ($p = 0.590$) (Fig. 3).

Adolescents

A total of 27 studies with a sample size of 49,370 participants were included in the analysis of social anxiety disorder prevalence among adolescents. Similar to the analysis for children, high heterogeneity was observed ($I^2 = 99.09$), leading to the application of the random-effects model. The pooled global prevalence of social anxiety disorders in adolescents was estimated at 8.3 (95% CI 5.7–11.8) (Fig. 4). The Egger test ($p = 0.792$) provided no evidence of publication bias (Fig. 5).

Table 2 Summary of characteristics of included studies of prevalence of social anxiety disorder in adolescents

References	Location	Sample	SAD prevalence (%)	Age	Instrument
Madasu et al. (2019a)	India	678	14.10	10–19	MINI KID
Mörtberg et al. (2022)	Sweden	2523	14.30	13–14	SPSQ-C[a]
Al-Yateem et al. (2020)	United Arab Emirates	968	20	13–18	SCARED[b]
Mpairwe et al. (2021)	Uganda	89	14.60	12–17	Y1-4R[c]
Gren-Landell et al. (2009)	Sweden	2128	4.40	12–14	SPSQ-C
Carlén et al. (2022)	Finland	504	19	18	DAWBA
Yuvaraj et al. (2021)	South India	1018	22.90	10–16	SPIN[d]
Abbo et al. (2013)	Uganda	885	5.20	10–19	M.I.N.I.-KID
Adewuya et al. (2007)	Nigeria	1090	4.80	13–18	MINI-Kid
Bener et al. (2011)	Qatar	1093	14.20	12–18	Anonymous questionnaire and diagnoses with psychiatrists
Madasu et al. (2019b)	India	678	14.30	10–19	SCARED
Dodangi et al. (2014)	Iran	226	4.90	12–18	K-SADS-PL
Nakie et al. (2022)	Northwest Ethiopia	876	40.20	15–25	SPIN
Gau et al. (2005)	Taiwan	1070	3.40	13–15	K-SADS-E
Shafi et al. (2023)	India	349	2.20	13–19	MINI PLUS AND MINI KID
Altwaijri et al. (2023)	Saudi Arabia	926	7.63	15–22	CIDI[e]
Ernst et al. (2022)	Germany	635	5.0	14–17	DIA-X-5/D-CIDI[f]
Arumuganathan et al. (2021)	India	31	10	14.85 ± 1.76	MINI-KID
Mallik and Radwan (2018)	Bangladesh	315	3.20	14–17	DAWBA
Nair et al. (2013)	India	500	2.20	11–19	K-SADS-PL
Nair et al. (2013)	India	500	20.20	11–19	SCARED
Capriola et al. (2017)	USA	20	45.0	14.62 ± 1.69	BFNE[g]
Waghachavare et al. (2014)	India	997	19.40	17.3	SPIN

Table 2 (continued)

References	Location	Sample	SAD prevalence (%)	Age	Instrument
Gren-Landell et al. (2009)	Sweden	3211	10.60	17.3 ± 0.64	SPSQ-C
Burstein et al. (2011)	US	5374	5.20	13–18	Structured interview
Spence et al. (2018)	Australia	2976	3.40	12–17	DISC-IV
Mohammadi et al. (2020)	Iran	19,710	2.10	10–18	K-SADS-PL

[A]the Social phobia screening questionnaire for children

[B]the Screen for child anxiety related disorders

[C]the Youth's inventory-4R

[d]The Social phobia inventory

[e]Composite international diagnostic interview

[f]Diagnostic interview

[g]Brief fear of negative evaluation questionnaire

Table 3 Summary of characteristics of included studies of prevalence of social anxiety disorder in youth

References	Location	Sample	SAD prevalence (%)	Age	Instrument
Jefferies and Ungar (2020)	Brazil, China, Indonesia, Russia, Thailand, US, Vietnam	6825	36.20	16–29	SIAS[a]
Mondin et al. (2013)	Brazil	1560	4.0	18–24	MINI
Locker et al. (2001)	New Zealand	705	12.10	18	DIS[b]
Altwaijri et al. (2023)	Saudi Arabia	885	6.16	23–30	CIDI
Ernst et al. (2022)	Germany	545	9.20	18–21	DIA-X-5/D-CIDI
Vriends et al. (2011)	Germany	1238	5.50	18–24	F-DIPS[c]
Uzonwanne (2014)	Nigeria	400	95.60	37.95 ± 12.19 (female) 37.35 ± 11.88 (male)	SPIN
Mick et al. (1998)	USA	76	13.16	18.92 ± 1.23	PASPQ[d]

[a]The social interaction anxiety scale

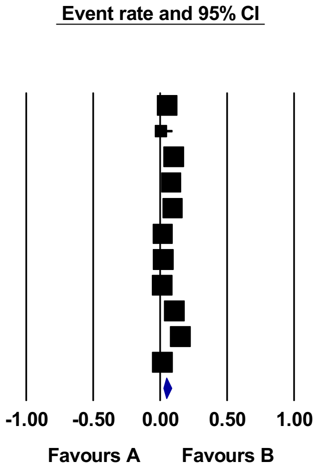
[b]The diagnostic interview schedule

[c]Diagnostisches interview fur psychische storungen

[d]Panic, anxiety, and social phobia questionnaire

Meta Analysis

Study name	Event rate	Lower limit	Upper limit	Z-Value	p-Value
Abbo et al	0.053	0.038	0.072	17.103-	0.000
Parvaresh et al	0.006	0.000	0.091	3.582-	0.000
Bener et al	0.102	0.080	0.128	16.263-	0.000
Dodangi et al	0.083	0.048	0.140	7.980-	0.000
Iverach et al	0.093	0.056	0.151	8.100-	0.000
Ezpeleta et al	0.019	0.011	0.034	13.477-	0.000
Van Roy et al	0.023	0.021	0.026	67.906-	0.000
Spence et al	0.016	0.012	0.020	29.656-	0.000
Paulus et al	0.107	0.091	0.124	24.035-	0.000
Hussong et al	0.151	0.131	0.174	20.048-	0.000
Mohammadi et al	0.017	0.015	0.020	52.999-	0.000
	0.047	0.025	0.084	9.386-	0.000



Meta Analysis

Fig. 2 Forest plot of the global prevalence of social anxiety disorder in children based on random effects method

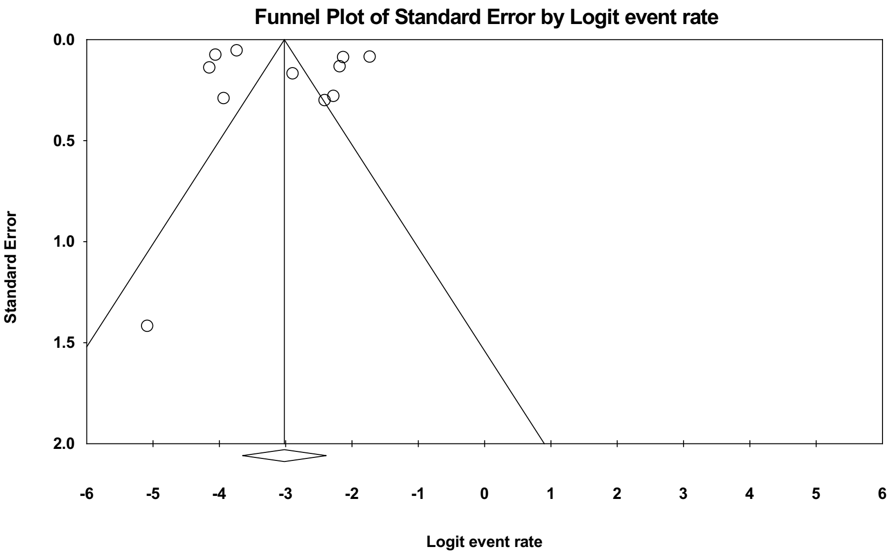


Fig. 3 Funnel plot of the publication bias in the reviewed studies

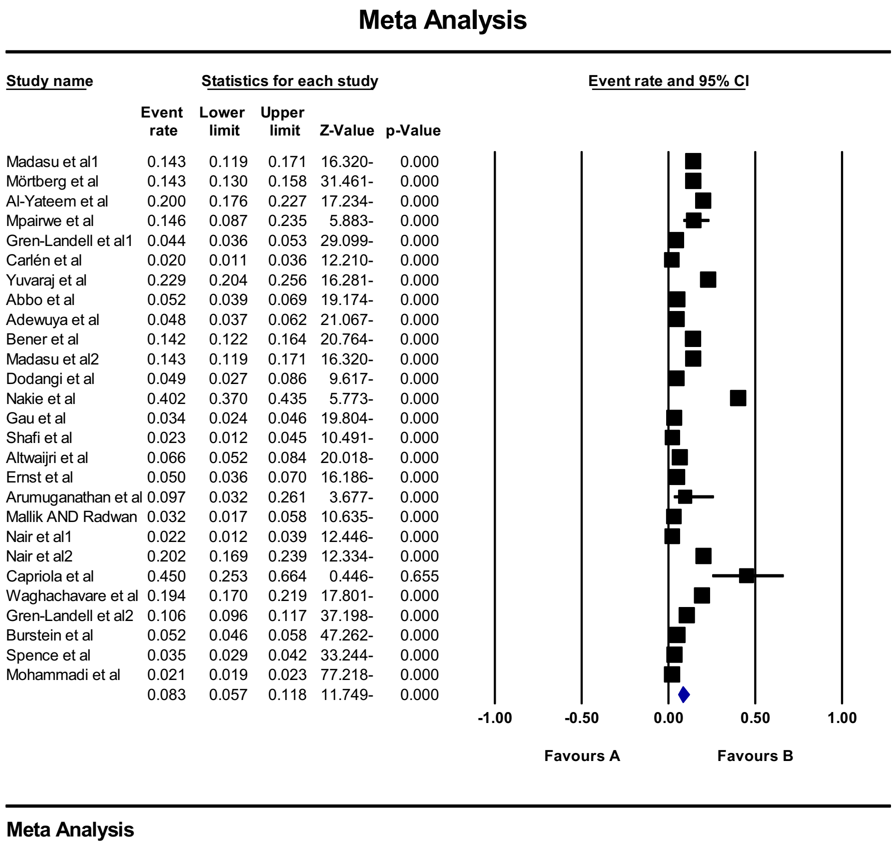


Fig. 4 Forest plot of the global prevalence of social anxiety disorder in adolescents based on random-effects method

Youth

Eight studies with a sample size of 12,234 participants were included in the analysis of social anxiety disorder prevalence among youth. Substantial heterogeneity was identified ($I^2 = 99.4$), justifying the use of the random-effects model. The pooled global prevalence of social anxiety disorders in young adults was estimated to be 17% (95% CI 6.5–37.6) (Fig. 6). The Egger test ($p = 0.145$) suggested no publication bias (Fig. 7).

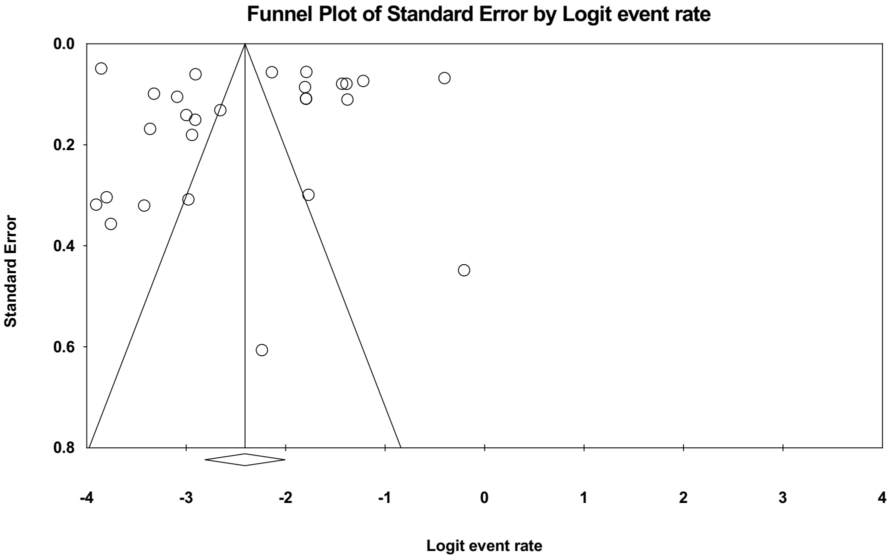


Fig. 5 Funnel plot of the publication bias in the reviewed studies

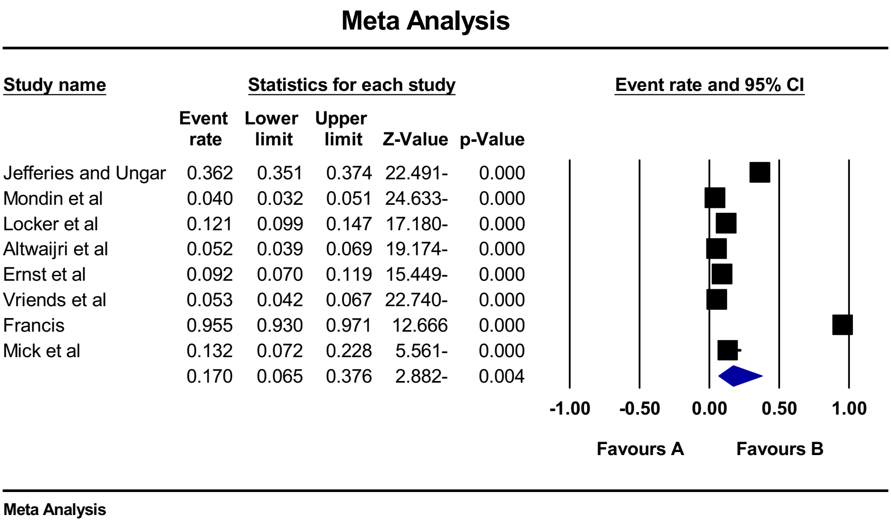


Fig. 6 Forest plot of the global prevalence of social anxiety disorder in youth based on random effects method

Discussion

This study investigated the global prevalence of social anxiety disorder across three age groups: children, adolescents and youth. The findings revealed an overall prevalence of 4.7%, 8.3% among adolescents and 17% among youth.

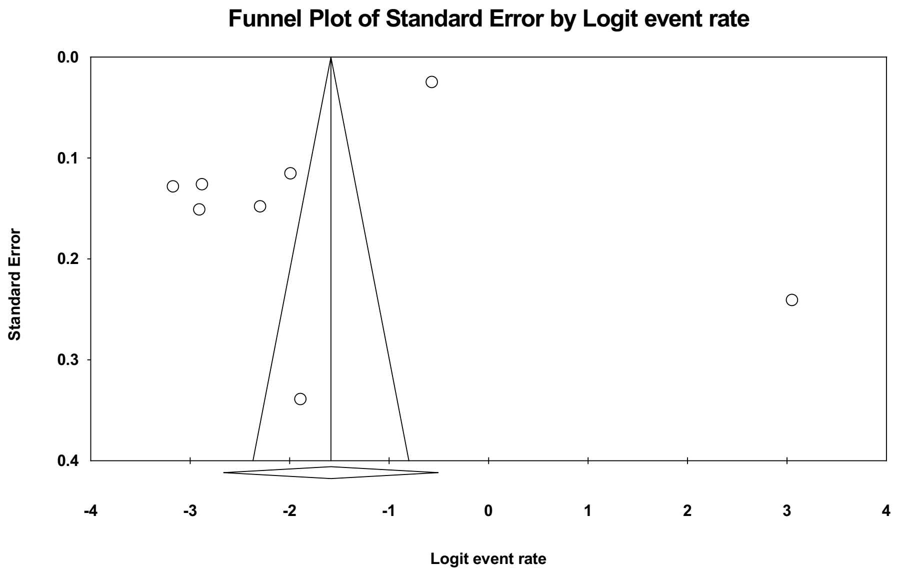


Fig. 7 Funnel plot of the publication bias in the reviewed studies

Social anxiety is recognized as a common mental health disorder, with potential onset in adolescence, although it can also manifest in childhood (Chapman et al., 2020; Esbjørn et al., 2021; Taylor et al., 2021). This chronic condition is known to have a lower recovery rate compared to other anxiety disorders and can be more resistant to treatment (Esbjørn et al., 2021; Taylor et al., 2021). Notably, the increase in the use of social media and the decline in face-to-face communication in recent years have been linked to an increase in social anxiety (Jefferies & Ungar, 2020).

The findings of this study align with previous studies reporting the prevalence of social anxiety disorder among children or adolescents (Abbo et al., 2013; Mohammadi et al., 2020; Spence et al., 2018). However, variations exist across populations. Mohammadi et al. (2020) reported a lower prevalence of 1.8% in Iranian children and adolescents. Other studies, such as studies conducted in Sweden and India, reported higher prevalence rates of 10.60% among Swedish adolescents and 10% among Indian adolescents (Arumuganathan et al., 2021; Gren-Landell et al., 2009). Similarly, Van Roy et al. (2009) estimated a prevalence of 2.30% in Norwegian children aged 8–13 years. These variations highlight the influence of sociodemographic factors. Studies suggest that factors such as rural residence, chronic illness, female gender, and family history can contribute to the development of social anxiety disorder (Jefferies & Ungar, 2020; Nakie et al., 2022). The high prevalence observed in Ethiopian adolescents (40.20%) might be partially explained by gender distribution, with 55.4% of participants being female (Capriola et al., 2017).

Cultural factors also likely play a role. The higher prevalence rates reported in some Western countries compared to others suggest a potential impact of ethnicity and cultural norms (Mohammadi et al., 2020). For instance, the 45% prevalence rate among teenagers in the United States might be influenced by its racial and ethnic diversity (Nakie et al., 2022).

Childhood and adolescent social anxiety can increase the risk of developing mental disorders in adulthood, as many adults with social anxiety disorder report experiencing early symptoms during adolescence (Chapman et al., 2020; Nakie et al., 2022). While social anxiety typically starts in adolescence, young adults often exhibit fear of social situations and avoidance behaviours (Ernst et al., 2022; Pearcey et al., 2021).

Several studies have reported on the prevalence of social anxiety among young adults in both the general population and those with specific conditions (Altwaijri et al., 2023; Ernst et al., 2022; McAllister et al., 2015). According to Lee and Clark's theory, increased self-awareness, decreased family interaction, and increased peer group interaction during adolescence can contribute to the development of social anxiety and vulnerability in young adults (Jefferies & Ungar, 2020).

In the present study, the prevalence of social anxiety among young adults is 17%. Jeffries and Yungar (2020) reported a higher prevalence of 36.20% among young people aged 16 to 29. In contrast, studies in New Zealand and Germany reported lower rates of 12.10% and 5.50% respectively (Locker et al., 2001; Vriends et al., 2011). These variations are somewhat similar to those observed in adolescents. In a cross-sectional study conducted in Nigeria, a much higher prevalence of social anxiety disorder was reported among this group of young people (95.6%), with over half being female (Uzonwanne, 2014). This suggests that sampling bias, measurement tool differences, and sample size variations can significantly influence reported prevalence rates.

Hofmann et al.'s research highlights the influence of sociocultural factors on social anxiety disorder (Hofmann & Asnaani, 2010). Collectivism versus individualism, social norms, gender roles, and gender role identification can all impact how social anxiety manifests. The same behaviour might be perceived differently across cultures. Cultural syndromes can also lead to the expectation of embarrassment in certain situations, and the meaning and experience of social anxiety disorder symptoms can be influenced by various factors (Hofmann & Asnaani, 2010). Therefore, careful consideration of an individual's sociocultural background is crucial when assessing social behaviours and attitudes.

Limitations

This study has several limitations. First, due to the limited number of studies on the prevalence of social anxiety disorder, particularly in children and youth, it was necessary to separate the prevalence rates by age range in some studies. However, the variation in age ranges considered for children, adolescents and youth across these studies introduced heterogeneity into the analysis. Secondly, the exclusion of non-English language studies and articles without age-stratified prevalence further limited the generalizability of the findings. This is because excluding studies from other languages restricts the geographic scope of the analysis. Thirdly, the lack of studies from various countries limited the generalizability of the results. Ideally, a broader range of countries would have been represented to enhance the global applicability of the findings. In addition, heterogeneity in the measurement of anxiety across studies also contributed to the overall variation in the results. The heterogeneity arising from factors such as sample size variations, participant demographics and different age ranges, is a common challenge in this type of research. However, future studies can employ statistical methods to mitigate the impact of such heterogeneity.

Conclusion

In the present study, the global prevalence of social anxiety disorder among children, adolescents and youth was calculated to be 4.7%, 8.3% and 17% respectively. By highlighting the prevalence of social anxiety disorder within these age groups, this research can encourage the development of targeted interventions and educational programs. Educating children, adolescents, youth, and parents about social anxiety disorder can promote early identification and access to appropriate support services. Early intervention can significantly improve the prognosis for individuals with social anxiety disorder.

Data and materials availability

Datasets are available through the corresponding author upon reasonable request.

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Author Contributions NS and MM and FB and PH contributed to the design, MM statistical analysis, and participated in most of the study steps. MM and FB and PH and NA and MH and MA prepared the manuscript. All authors have read and approved the content of the manuscript.

Declarations

Conflict of Interest The authors declare that they have no conflict of interest.

Ethics Approval and Consent to Participate Ethics approval was received from the ethics committee of deputy of research and technology, Kermanshah University of Medical Sciences (IR.KUMS.REC.1402.374).

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